Red, Falling, Nearest Star: Does Exposure to Science Fiction Literature Predict Knowledge and Processing of Genre-specific Collocations?

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Introduction

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(1) sit in your chair *vs.* sit in your truck (Freq: high vs. low)

(Bannard & Matthews, 2008)

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(2) in one of the *vs.* in the context of the (Freq: high vs. low)

(3) a great deal of *vs.* is one of the (PMI: high vs. low)

(Ellis, Simpson-Vlach, & Maynard, 2008)

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(4) stinky baby *vs.* silly baby (Info(A): high vs. low)

(Bannard, Rosner, & Matthews, 2017)

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

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delicate tea
feeble tea
frail tea
powerless tea
weak tea

(6)
deliver a speech
hold a speech
perform a speech
present a speech
utter a speech

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Instead, task performance was significantly correlated with print exposure (Author Recognition Test (ART), Acheson, Wells, & MacDonald, 2008).

The Current Study

Question What is the relationship between print exposure, corpus-based measures of association, and language use and processing?

Method A bigram completion task was conducted to determine the effect of print exposure to English science fiction (SF) literature and corpus-based measures of association on readers' responses.

Results Responses of more experienced SF readers with higher print exposure mirrored the usage distribution of bigrams in SF literature more closely.

Overview

1. Introduction

2. Method

3. Preliminary Results

4. Discussion

Section 2

Method

The Science Fiction Corpus (SFC)

Corpus

- 1,548 English science fiction texts from Gutenberg.org
- 32,402,850 words, 0.01 type/token ratio, 16.49 token/sentence ratio
- POS tagging with TreeTagger with BNC tagset for Linux

Rank	N	A	RS authors	RS titles
1	man	other	H.G. Wells	The Case of Summerfield
2	time	great	Zona Gale	The Infra-Medians
3	way	little	Murray Leinster	Postmark Ganymede
4	men	good	H. Beam Piper	The Calm Man
5	eyes	old	John Joseph McGuire	Perfect Control
6	hand	long	Ron Cocking	The Buttoned Sky
7	face	new	Fritz Leiber	Venus Boy
8	world	small	Henry Slesar	Syndrome Johnny
9	people	sure	C.L. Moore	Fifteen Hundred Miles An Hour
10	earth	right	Mike Lewis	Master of Life and Death

Corpus-based Measures

- 593,873 unique adjective-noun bigrams
- Joint frequency, mutual information, informativity of adjective

AN	Rank	BNC	PMI(AN)	BNC	Info(A)
old man	1	21	4.40	4.34	4.09
young man	2	18	4.98	5.01	4.23
long time	3	4	4.15	4.43	4.53
other side	4	12	4.57	4.29	3.32
extensive research	5	3890	9.23	4.33	1.01
other hand	6	2	3.77	4.87	4.47
solar system	7	1573	7.64	5.58	2.53
great deal	8	6	6.06	6.89	1.26
human beings	9	49	7.16	8.74	1.16
whole thing	10	74	4.75	5.06	4.71

Bigram Completion Task

- 3,398 genre-specific bigrams (collocations)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

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nearest

Author Recognition Test (Print Exposure)

- Author recognition test to assess participants' SF print exposure (Acheson et al., 2008; Stanovich & West, 1989)
- 16 SFC author names, 16 fake names (Martin-Chang & Gould, 2008)
- Participants indicate authors they know, penalty for fakes

Please click on the authors you recognize. Careful! I have added some fake names to make sure you are not just guessing.

Alex James	☐	...	☐
Andrew Lang	☐	Richard Wilson	☐
Arthur Conan Doyle	☐	Robert Louis Stevenson	☐
Ayn Rand	☐	Rudyard Kipling	☐
...	☐	Thomas Bever	☐

Procedure and Predictions

Procedure

- Ibex farm (Drummond, 2013)
- Participant recruitment online (callforparticipants.com, PRO, SFWA, r/SampleSize)
- Instruction $\rightarrow$ ART $\rightarrow$ 3 practice trials $\rightarrow$ BCT $\rightarrow$ Lottery
- Each participant completed random sample of 50 nouns
- Response (adjective) and response time (RT) were recorded

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Predictions

- Participants are sensitive to usage distributions.
- More experienced SF readers produce more strongly associated and more informative bigrams.
- More experienced SF readers produce more genre-specific bigrams (collocations).
- RT decreases with increasing association strength.

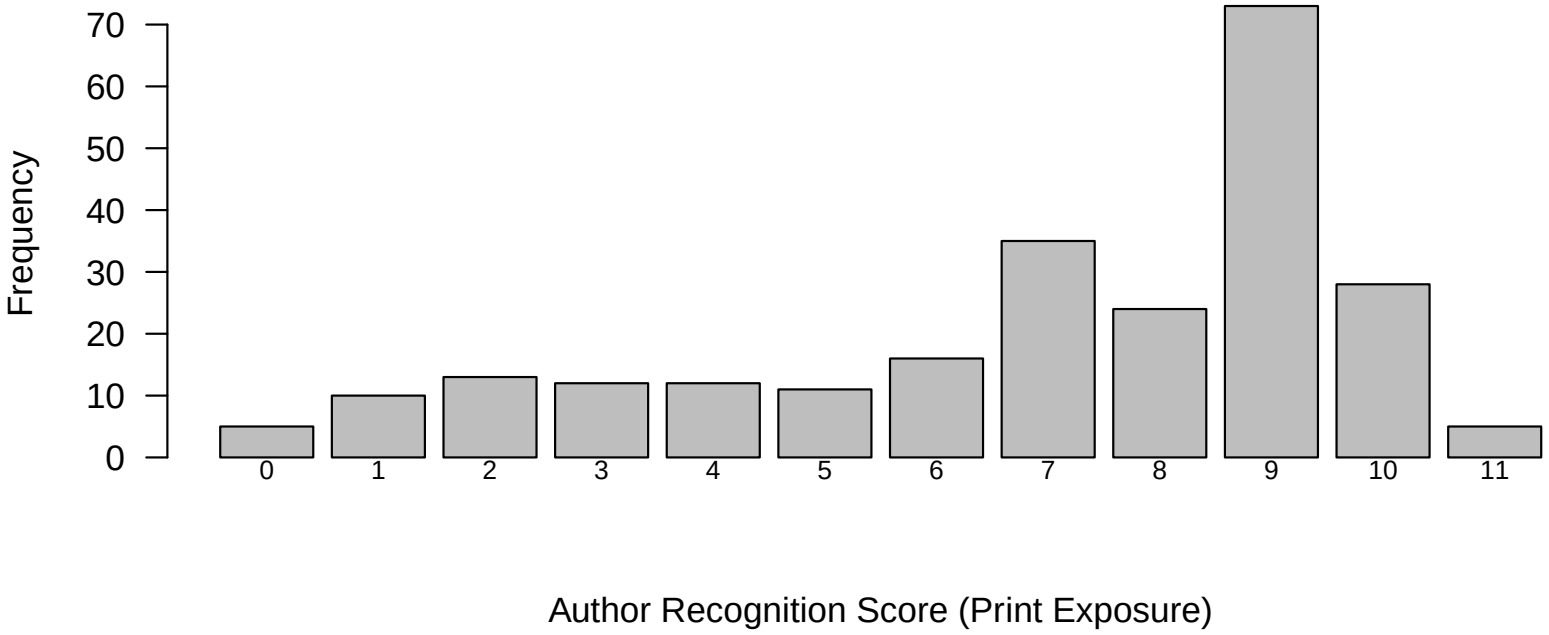
Section 3

Preliminary Results

Preliminary Results (11/12/2018 – 01/22/2019)

Table: Summary of participants' background information

Variable	
N	244
Sex (female): %	59.02
Age in years: *M* (*SD*)	38.38 (15.33)
L1	English
ART score (-16 – 16): *M* (*SD*)	6.97 (2.82)



Joint Frequency Distribution

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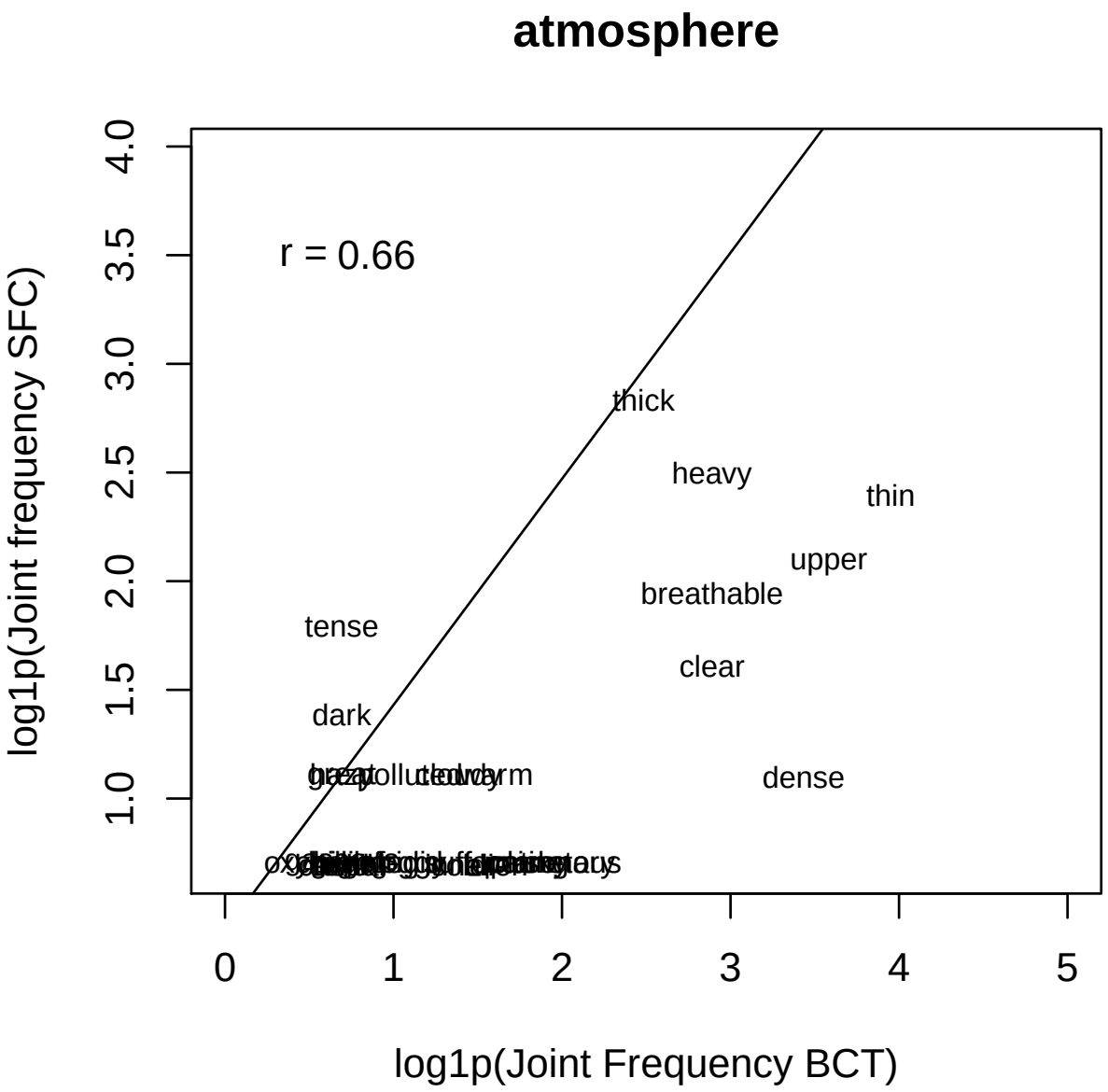
Joint frequency bigrams BCT $\sim$ Joint frequency bigrams SFC/BNC

(Ellis, Römer, & O'Donnell, 2016, pp. 103–107)

Joint Frequency Distribution



Joint Frequency Distribution

N	n(A)	r SFC	p
arm	18	0.82	***
safety	13	0.69	**
direction	22	0.60	**
war	29	0.51	**
cars	14	0.42	.
voice	42	0.37	**
circumstances	30	0.33	*
condition	28	0.25	ns
building	27	0.16	ns
star	21	0.03	ns

M of r (n of $p \leq .05$): 0.37 (56)

Joint Frequency Distribution

N	n(A)	r SFC	p	N	n(A)	r BNC	p
arm	18	0.82	***	money	12	0.86	***
safety	13	0.69	**	direction	22	0.80	***
direction	22	0.60	**	appearance	29	0.72	***
war	29	0.51	**	floor	24	0.67	***
cars	14	0.42	.	orders	31	0.63	***
voice	42	0.37	**	group	25	0.59	**
circumstances	30	0.33	*	office	29	0.52	**
condition	28	0.25	ns	star	21	0.44	*
building	27	0.16	ns	state	36	0.38	*
star	21	0.03	ns	works	13	0.25	ns

M of r (n of $p \leq .05$): 0.37 (56) M of r (n of $p \leq .05$): 0.55 (81)

PMI(AN) and Info(A) Distribution

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- More experienced SF readers produce more strongly associated and more informative bigrams.

PMI(AN) and Info(A) Distribution

Prediction

- PMI(AN), Info(A) $\sim$ print exposure (ART score)

PMI(AN) and Info(A) Distribution

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ID	AN	ART_SCORE	FreqAN_SFC	PMI_SFC	InfoA_SFC
1	hidden meaning	10	12	5.35	7.19
14	poor health	8	5	3.30	7.98
34	final aspect	7	1	2.74	9.77

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PMI(AN) and Info(A) Distribution

Prediction

- value $\sim$ print exposure $\times$ SFC measures (Freq/PMI/Info)

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PMI(AN) and Info(A) Distribution

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- value $\sim$ print exposure $\times$ SFC measures (Freq/PMI/Info)

	b	*SE*	*t*	*p*
Intercept	2.21	0.06	34.35	***
Print exposure (ART score)	0.01	0.01	1.11	ns
PMI(AN)	0.87	0.09	10.2	***
Info(A)	6.67	0.09	78.47	***
Print exposure x PMI(AN)	0.08	0.01	6.83	***
Print exposure x Info(A)	-0.03	0.01	-2.96	**

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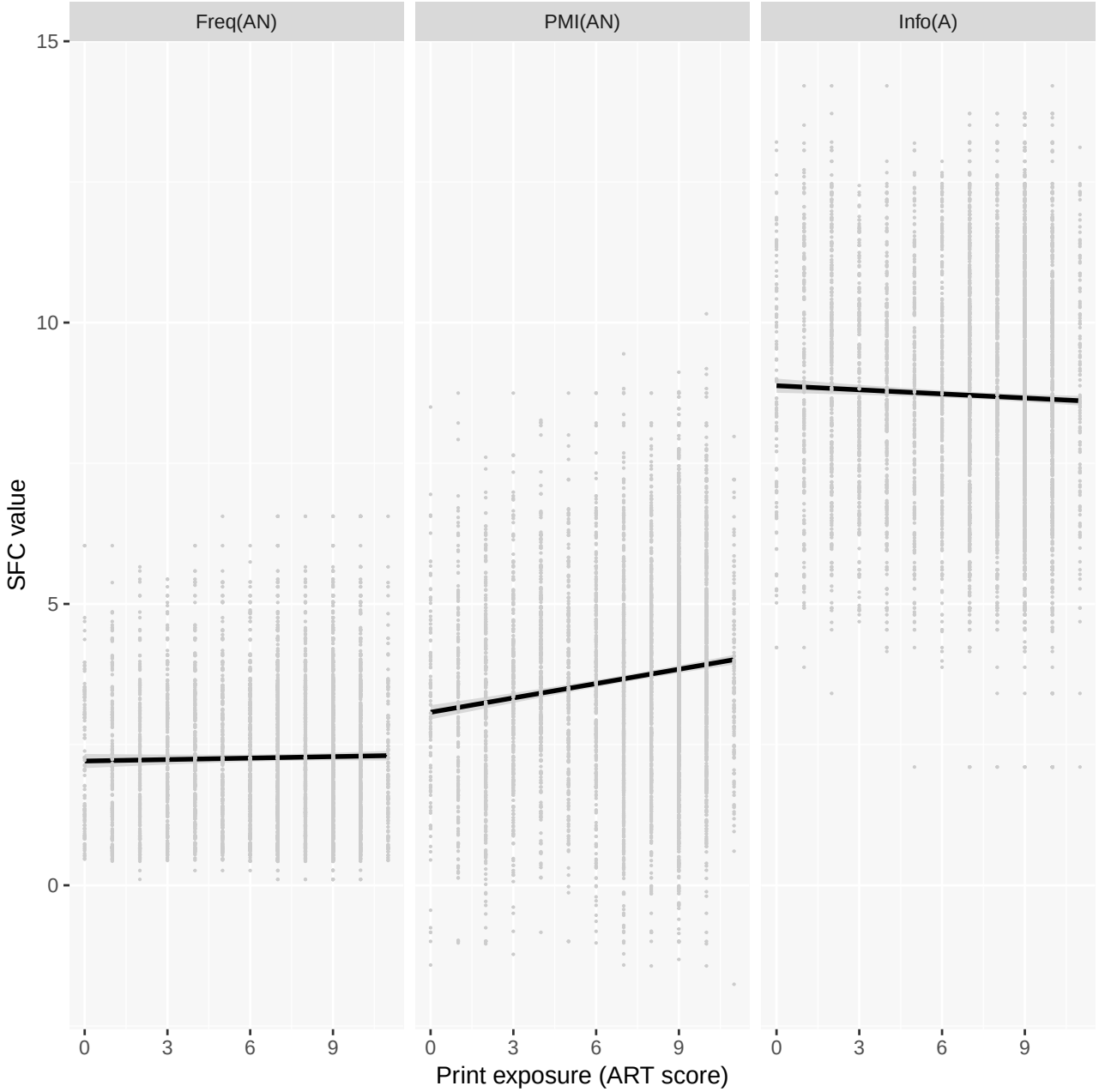

Genre-specific Bigrams

Prediction

- More experienced SF readers produce more genre-specific bigrams.

ID	AN	ART_SCORE	PMI_SFC	PMI_BNC
1	high council	10	3.90	1.18
1	hidden meaning	10	5.35	5.23
2	terrible state	7	2.67	3.42
1	deep state	10	0.76	-0.68

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ID	AN	ART_SCORE	PMI_SFC	PMI_BNC	SFColl
1	high council	10	3.90	1.18	SFColl
1	hidden meaning	10	5.35	5.23	NotSFColl
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Genre-specific Bigrams

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- SFColl(yes/no) $\sim$ print exposure (ART score)

ID	AN	ART_SCORE	PMI_SFC	PMI_BNC	SFColl
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	b	*SE*	*z*	*p*
Intercept	-2.65	0.12	-21.42	***
Print exposure (ART score)	0.04	0.01	3.16	**

Response Time

Prediction

- RT decreases with increasing association strength.

Response Time

Prediction

- $\text{RT} \sim \text{Freq(AN)} + \text{PMI(AN)} + \text{Info(A)} + \text{Trial}$

Response Time

Prediction

- RT $\sim$ Freq(AN) + PMI(AN) + Info(A) + Trial

	b	*SE*	*t*	*p*
Intercept	2390.97	541.47	4.42	***
log1p(Freq(AN))	-29.42	65.17	-0.45	ns
PMI(AN)	-131.59	26.32	-5	***
Info(A)	-30.97	41.46	-0.75	ns
Trial	-1.15	2.53	-0.45	ns

Section 4

Discussion

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Results

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Language learning involves implicit sequential learning, similar to learning in recurrent neural networks (Elman, 1990).

(Diessel, 2016; Ellis et al., 2016; Schmid, 2018; Sinclair, 1991)

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More experienced language users rely on more fine-grained and context-sensitive analysis of their input.

(Durrant & Schmitt, 2009; Ellis et al., 2008; Jach, 2018a, 2018b; Schmitt, 2012)

Limitations and Outlook

Limitations

- Mismatch between BCT and SFC bigrams
- No effect of informativity on responses, probably due to task
- No print exposure $\times$ mutual information in processing

Outlook

- Bi-, tri-, quadro-, n-grams
- Links to rhetorical genre analysis

Thank you for your attention.

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